Wish Upon a Star 🌠

Estimated time: 30-40 minutes
Learning Goals: Practice OOP core concepts such as abstract classes, interfaces, inheritance through a creative Java project

Story Background

Somewhere in the universe, there are stars either ancient, powerful, or completely unhinged. Every night, people have a chance to send up a wish. However, the plot twist is that these stars are completely unpredictable. Only one random star hears each wish and people do not know ahead which one will hear their wish. It decides on its own whether to grant the wish or not.

ShootingStar - always tries their best to grant it but misreads the wish
DyingStar - too tired of this; there's 50/50 chance it'll be bothered enough to grant the wish
BabyStar - always grants something but never what you asked for

Your task is to write a program that asks the user for their name, and wish. Then, randomly pick a star to respond.

What You'll Build:

1) Main.java - interactive menu program
2) Wish.java - interface (provided)
3) Star.java - abstract class (provided)
4) ShootingStar.java - you need to implement this subclass
5) DyingStar.java - you need to implement this subclass
6) BabyStar.java - you need to implement this subclass
7) YourStar.java - your own creative star! (You may give the name you want. Doesn't have to be YourStar.java)

Part 1: Understand the given code and design your custom star (5-10 mins)

Wish.java and Star.java are already given in the repository. You do not need to modify them. Simply add them in your directory.

Wish.java

```
package main;

public interface Wish {
    String getWisher();
    String getDesire();

    default String fullWish() {
        return getWisher() + " wishes for " + getDesire();
    }
}
```

Star.java

```
package main;

public abstract class Star {
    protected String name;
    protected String mood;

    public Star(String name, String mood) {
        this.name = name;
        this.mood = mood;
    }

    // You need to decide which subclass does what with the wish
    public abstract void grantWish(Wish wish);

    public void introduce() {
        System.out.println("Star " + name + " is here, feeling " + mood +
"!");
    }
}
```

App.java

This is (as usual) your program's entry point. A basic structure to help you get started. You do not need to strictly follow this structure.

```
package main;

import java.util.Scanner;
import java.util.Random;

public class App
{
    public static void main( String[] args )
    {
        Scanner scanner = new Scanner(System.in);
        Random random = new Random();

        /*
        Create some stars here. For example:

        Stars[] stars = {
            new ShootingStar("name1", "mood1"),
            new DyingStar("name2", "mood2"),
            ...
        };
        */

        System.out.println("==========================================");
        System.out.println("** WISH UPON A STAR **");
        System.out.println("==========================================");

        while (true) {

            /*
            TO DO: Make a menu for the user to choose a star to wish upon,
and to input their wish.

            For example:

            // A bunch of code to print the menu
```

```
            String choice = scanner.nextLine().trim();

            case "1" -> {
                // Ask for user's name
                String wisher = scanner.nextLine().trim();

                // Ask for user's desire
                String desire = scanner.nextLine().trim();

                // TO DO: create a Wish object using the user's input
                // return wisher and desire

                }
            */

                //  pick a random star
                Star randomStar = stars[random.nextInt(stars.length)];

                // grant the wish using the random star
                // Add some whimsical output here to make it more fun
                randomStar.introduce();
                randomStar.grantWish(wish);

            /*

            case "2" -> {
                // Exit the program
                System.out.println("Farewell, dearest stargazer!");
                break;
            }
            */


        }
        scanner.close();
    }
}
```

Part 2: Implement the Stars

Each subclass must extend Star.jav, call super(name, mood) in the constructor, and override grantWish(Wish wish). Each star has a different attitude toward granting wishes.

ShootingStar - always tries their best to grant it but misreads the wish

ShootingStar travels fast! Tries to grant the wish but it mishears it all the time. The output should make it clear what the user's wish was and what ridiculous thing it actually granted instead.

Example output:

```
Your name: Clare
Your wish: great power

A star stirs in the darkness . . .
=====================================================
StarDash appears feeling excited!
ShootingStar zooms in and wish is received!
Clare wants ... great power ...
Granting: great flower. Proud of myself and you're welcome!
=======================================================
```

```java
package main;

public class ShootingStar extends Star {
    public ShootingStar(String name, String mood) {
        super(name, mood);
    }

    @Override
    public void grantWish(Wish wish) {
       // TO DO: always grant the wish, but also mishears the wish
       // use wish.getWisher() and wish.getDesire()
    }
}
```

DyingStar - may or may not grant the wish

DyingStar is ancient and already exhausted. Use Math.random() (or a Random object) to give it a 50% chance of granting the wish. The twist is that if it does grant it, it complains. If it doesn't, it gives a dramatic excuse.

Example output - granted:

```
A star stirs in the darkness . . .

==================================================
Grumbie appears feeling exhausted . . .
DyingStar sighs ...
Fine ... you’re lucky I’m feeling generous today.
Clare wants a pet dragon. I suppose I’ll manage.
Wish granted. Now, my light fades a little more. Hope you’re happy.
====================================================
```

Example output - refused:

```
A star stirs in the darkness . . .

==================================================
Grumbie appears feeling exhausted . . .
DyingStar sighs ...
A pet dragon . . . how cute. I’m (not) sorry but nope.
I’m literally on the brink of entirely collapsing. Come back never.
====================================================
```

Hint: use Math.random() <0.5 to get 50/50 chance inside method grantWish()

BabyStar - always grants but never correctly

BabyStar is very young and enthusiastic. It’s always excited to make people’s wishes come true but it has no clear idea of what a wish actually is. It should grant something completely random (unrelated), with max excitement.

Example output:

```
==================================================
Twinkle appears feeling overwhelmed but excited. . .
WOW!! I finally caught a wish! Let’s see .... Clare wants a pet dragon!!!
```

```
I don't know exactly what a pet dragon is but I can try!
I have granted Clare a pet stone!
I hope they like it because I worked very hard on it.
=====================================================
```

Your Custom Star

- Extend Star.java and call super(name,mood) in the constructor
- Override grantWish (Wish wish) and use both getWisher() and getDesire()
- Have a clear, consistent personality with at least 2 lines of output
- Make its own decision about whether to grant the wish or not

Example Program Flow:

```
==========================================
 * WISH UPON A STAR *
==========================================

What would you like to do?
  1. Make a wish
  2. Quit
Enter choice: 1
Your name: Dreamer
Your wish: a pet dragon

A star stirs in the darkness...
------------------------------------------
Star Twinkle is here, feeling overwhelmed but excited!
WOW!! I finally caught a wish! Let's see... Dreamer wants a pet dragon!!!
I don't know exactly whata pet dragon is, but I can try!
I have granted Dreamer... a pet stone.
I hope they like it because I worked very hard on it.
------------------------------------------

What would you like to do?
  1. Make a wish
  2. Quit
Enter choice: 2
Farewell, dearest stargazer!
```